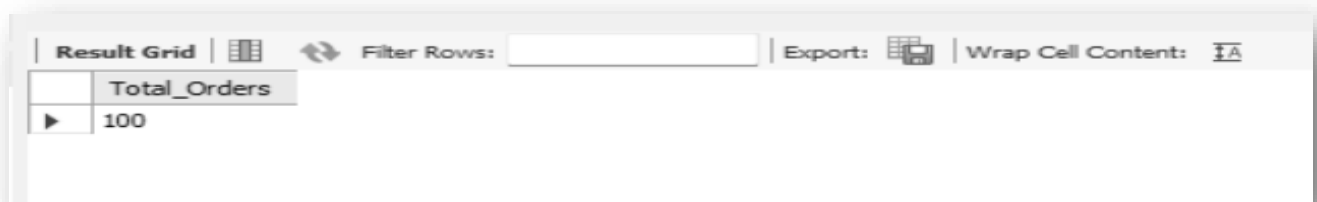


Supply Chain Data Analysis Practical Report

1. Find total orders by region.

```
select count (OrderID) AS Total _Orders  
from supply_chain;
```

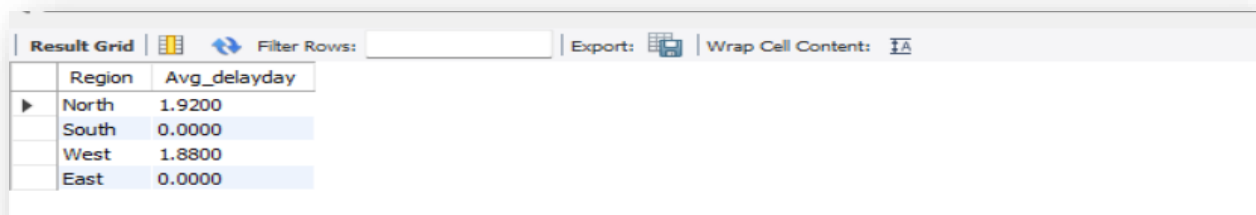


The screenshot shows a database interface with a 'Result Grid' tab. The grid contains one row with the column header 'Total_Orders' and the value '100'. Above the grid, there are controls for 'Filter Rows', 'Export', and 'Wrap Cell Content'.

Total_Orders
100

2. Find average delay days by region.

```
Select Region, avg(DelayDays) AS Avg_delayday  
from supply_chain  
group by Region;
```



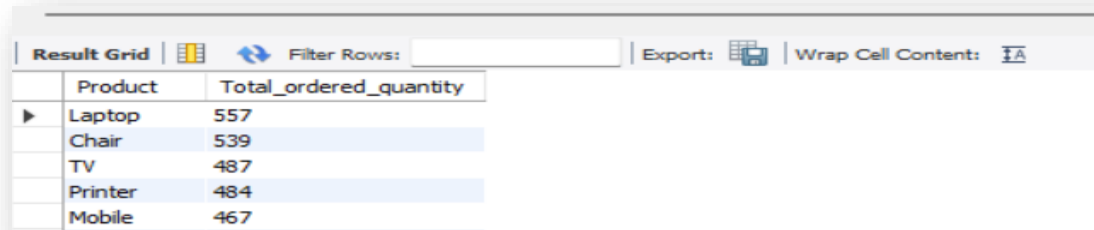
The screenshot shows a database interface with a 'Result Grid' tab. The grid contains four rows with columns 'Region' and 'Avg_delayday'. The rows are: North (1.9200), South (0.0000), West (1.8800), and East (0.0000). Above the grid, there are controls for 'Filter Rows', 'Export', and 'Wrap Cell Content'.

Region	Avg_delayday
North	1.9200
South	0.0000
West	1.8800
East	0.0000

3. Find total ordered quantity by product.

```
select d.Product,  
sum(s.OrderedQty) AS Total_ordered_quantity  
from supply_chain s  
join dim_material d  
on s.ProductID = d.ProductID  
group by d.Product  
order by Total_ordered_quantity desc;
```

```
#order by desc;
```



A screenshot of a SQL query result grid. The grid has two columns: 'Product' and 'Total_ordered_quantity'. The data is as follows:

Product	Total_ordered_quantity
Laptop	557
Chair	539
TV	487
Printer	484
Mobile	467

4. JOIN the two tables and display Product, Supplier and OrderedQty.

```
select d.Product, d.Supplier, s.OrderedQty
from supply_chain s
join dim_material d
on s.ProductID = d.ProductID;
```



A screenshot of a SQL query result grid. The grid has three columns: 'Product', 'Supplier', and 'OrderedQty'. The data is as follows:

Product	Supplier	OrderedQty
Laptop	Dell	45
Chair	IKEA	57
Mobile	Samsung	7
Printer	HP	48
Table	HomeCenter	14

5. Find total sales value by product.

```
select d.Product,
sum(s.OrderedQty * d.`Unit Price`) AS Total_sales_value
from supply_chain s
join dim_material d
on s.ProductID = d.ProductID
group by d.Product
order by Total_sales_value desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Product	Total_sales_value			
▶	Laptop	27850000			
	TV	21915000			
	Mobile	14010000			
	Printer	7260000			
	Table	4152000			

6. Find which supplier has the highest number of orders.

```

select d.Supplier, count(s.OrderID) AS Total_Orders
from supply_chain s
join dim_material d
on s.ProductID = d.ProductID
group by d.Supplier
order by Total_orders desc;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Supplier	Total_Orders			
▶	Dell	15			
	IKEA	15			
	Samsung	14			
	HP	14			
	HomeCenter	14			

7. Find the product with the highest ordered quantity.

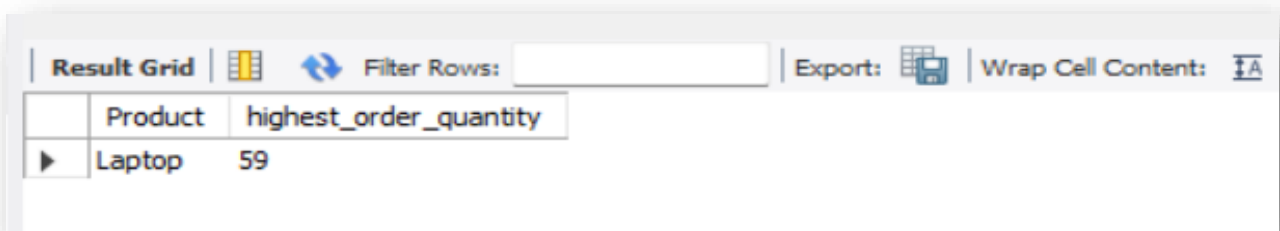
```

select d.Product
, max(s.OrderedQty) AS highest_order_quantity
from supply_chain s
join dim_material d
on s.ProductID = d.ProductID
group by d.Product

```

order by highest_order_quantity desc

limit 1;



The screenshot shows a database interface with a 'Result Grid' tab. The grid has two columns: 'Product' and 'highest_order_quantity'. The first row shows 'Laptop' with a value of 59. The interface includes a 'Filter Rows' search bar, an 'Export' button, and a 'Wrap Cell Content' toggle.

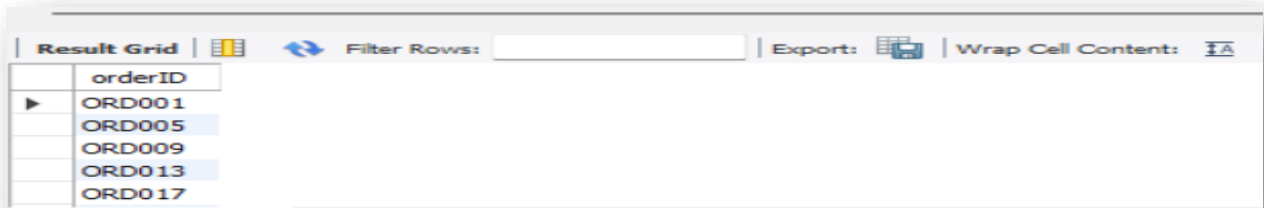
	Product	highest_order_quantity
▶	Laptop	59

8. Display all orders from the North region.

select orderID

from supply_chain

where Region = 'North';



The screenshot shows a database interface with a 'Result Grid' tab. The grid has one column: 'orderID'. The first five rows show 'ORD001', 'ORD005', 'ORD009', 'ORD013', and 'ORD017'. The interface includes a 'Filter Rows' search bar, an 'Export' button, and a 'Wrap Cell Content' toggle.

	orderID
▶	ORD001
	ORD005
	ORD009
	ORD013
	ORD017

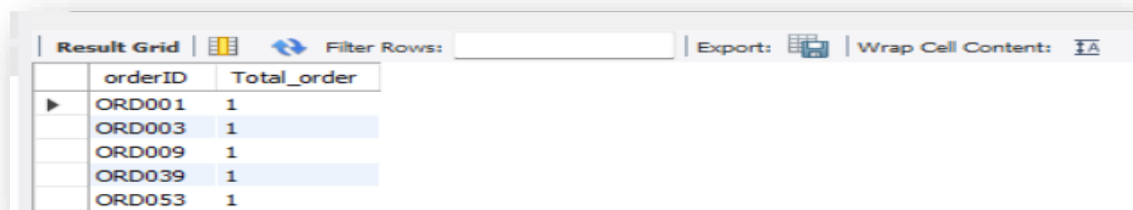
9. Find all orders where DelayDays is greater than 2.

select orderID,count(*) AS Total_order

from supply_chain

where DelayDays > 2

group by OrderID;

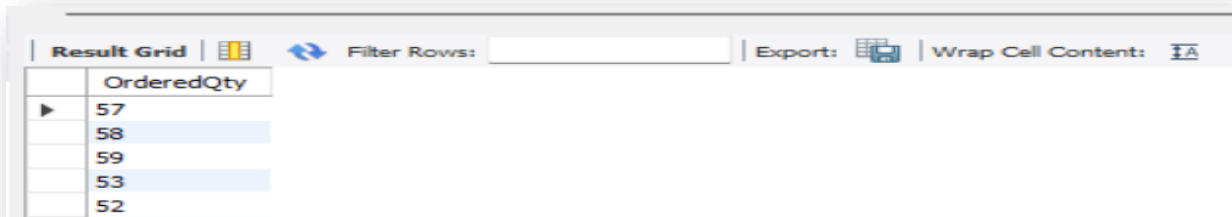


The screenshot shows a database interface with a 'Result Grid' tab. The grid has two columns: 'orderID' and 'Total_order'. The first five rows show 'ORD001', 'ORD003', 'ORD009', 'ORD039', and 'ORD053', each with a 'Total_order' of 1. The interface includes a 'Filter Rows' search bar, an 'Export' button, and a 'Wrap Cell Content' toggle.

	orderID	Total_order
▶	ORD001	1
	ORD003	1
	ORD009	1
	ORD039	1
	ORD053	1

10. Find all orders where OrderedQty is greater than 50.

```
select OrderedQty  
from supply_chain  
where OrderedQty > 50;
```

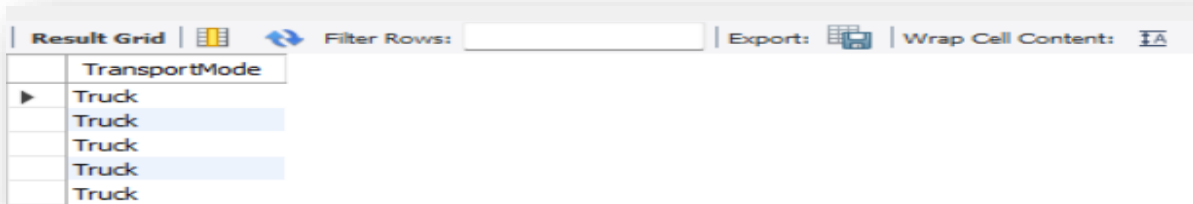


A screenshot of a SQL query result grid. The grid has a header row with the column name 'OrderedQty'. Below the header, there are five rows of data with values 57, 58, 59, 53, and 52. The grid is part of a software interface with a toolbar at the top containing icons for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'.

OrderedQty
57
58
59
53
52

11. Display orders transported by Truck.

```
select TransportMode  
from supply_chain  
where TransportMode = 'Truck';
```



A screenshot of a SQL query result grid. The grid has a header row with the column name 'TransportMode'. Below the header, there are five rows of data, all with the value 'Truck'. The grid is part of a software interface with a toolbar at the top containing icons for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'.

TransportMode
Truck
Truck
Truck
Truck
Truck

12. Find orders where DeliveredQty is less than OrderedQty.

```
select DeliveredQty , OrderedQty  
from supply_chain  
where DeliveredQty < OrderedQty;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	DeliveredQty	OrderedQty			
▶	5	7			
	45	48			
	22	24			
	43	45			
	28	30			

13. Find orders where OTIF = 0 and DelayDays > 0.

```
select OTIF, DelayDays
from supply_chain
where OTIF = 0 and DelayDays > 0;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	OTIF	DelayDays			
	0	3			
	0	3			
▶	0	2			
	0	3			
	0	2			

14. Find the total number of orders.

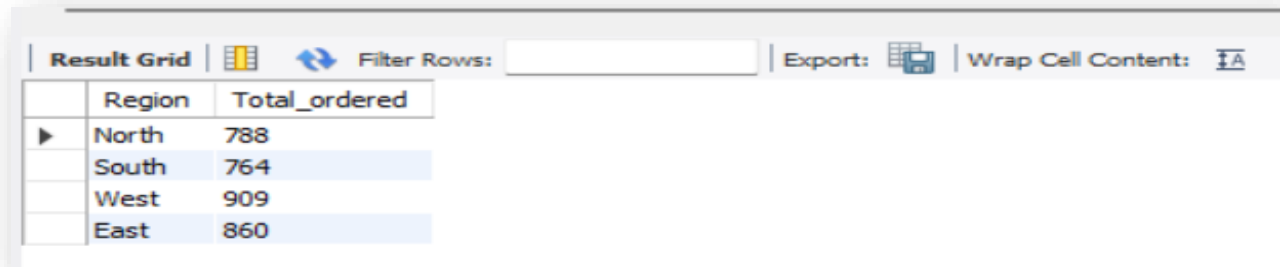
```
select count (OrderID) AS Total_number_Orders
from supply_chain;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Total_number_Orders			
▶	100			

15. Find the total ordered quantity by region.

```
select Region, Sum (OrderedQty) AS Total_ordered
from supply_chain
```

group by Region;

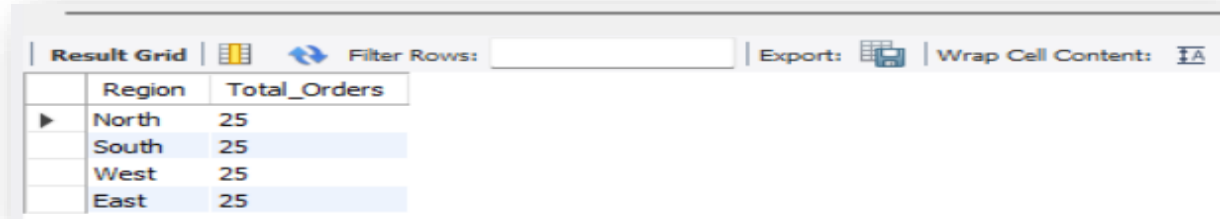


The screenshot shows a database query result grid with a toolbar at the top. The toolbar includes a 'Result Grid' button, a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' button. The table has two columns: 'Region' and 'Total_ordered'. The data is as follows:

	Region	Total_ordered
▶	North	788
	South	764
	West	909
	East	860

16. Find regions having more than 20 orders.

```
select Region, count (OrderID) AS Total_Orders  
from supply_chain  
group by Region  
having count(OrderID) > 20;
```



The screenshot shows a database query result grid with a toolbar at the top. The toolbar includes a 'Result Grid' button, a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' button. The table has two columns: 'Region' and 'Total_Orders'. The data is as follows:

	Region	Total_Orders
▶	North	25
	South	25
	West	25
	East	25

17. Find transport modes having an average delay greater than 2 days.

```
select TransportMode, avg(DeliveredQty) AS Average_deay  
from supply_chain  
group by TransportMode  
having avg(DeliveredQty) > 2;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	TransportMode	Average_deay			
▶	Truck	30.8125			
	Ship	32.0476			
	Air	34.1538			

18. Find the average unit price by product category.

```
select d.`Product Category`,avg(`Unit Price`) AS Average_UnitPrice
from supply_chain s
join dim_material d
on s.ProductID = d.ProductID
group by d.`Product Category`;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Product Category	Average_UnitPrice			
▶	Electronics	41860.4651			
	Furniture	9413.7931			
	Office Supplies	7750.0000			

19. Using an INNER JOIN, display Product, Supplier, and OrderedQty.

```
select d.Product,d.Supplier, s.OrderedQty
from supply_chain s
inner join dim_material d
on s.ProductID =d.ProductID;
```


Result Grid			
Filter Rows:			
Export: Export			
Wrap Cell Content: Wrap			
	Product	Supplier	OrderedQty
▶	Laptop	Dell	45
	Chair	IKEA	57
	Mobile	Samsung	7
	Printer	HP	48
	Table	HomeCenter	14

20. Find the total ordered quantity for each product using an INNER JOIN.

```

select d.Product, sum(s.OrderedQty) AS Total_OrderedQty
from supply_chain s
inner join dim_material d
on s.ProductID = d.ProductID
group by d.Product;

```

Result Grid		
Filter Rows:		
Export: Export		
Wrap Cell Content: Wrap		
	Product	Total_OrderedQty
▶	Laptop	557
	Chair	539
	Mobile	467
	Printer	484
	Table	346

21. Display all orders

with: OrderID, Product, Supplier, OrderedQty, DeliveredQty?

```

select s.OrderID,
d.Product,
d.Supplier,
s.OrderedQty,
s.DeliveredQty
from supply_chain s

```

```

left join dim_material d
on s.ProductID = d.ProductID;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

IA

	OrderID	Product	Supplier	OrderedQty	DeliveredQty
▶	ORD001	Laptop	Dell	45	45
	ORD002	Chair	IKEA	57	57
	ORD003	Mobile	Samsung	7	5
	ORD004	Printer	HP	48	45
	ORD005	Table	HomeCenter	14	14

22. Find the total ordered quantity by supplier using a LEFT JOIN.

```

select d.Supplier,sum(s.OrderedQty) Total_OrderedQty
from supply_chain s
left join dim_material d
on s.ProductID = d.ProductID
group by d.Supplier;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Supplier	Total_OrderedQty
▶	Dell	557
	IKEA	539
	Samsung	467
	HP	484
	HomeCenter	346

23. Using a RIGHT JOIN, display all products and their corresponding orders.

```

select d.ProductID,
d.Product,
d.Supplier,
s.OrderID,
s.OrderedQty,

```

```

s.DeliveredQty
from supply_chain s
right join dim_material d
on s.ProductID = d.ProductID;

```

Result Grid						
Filter Rows:						
Export:  Wrap Cell Content: 						
	ProductID	Product	Supplier	OrderID	OrderedQty	DeliveredQty
▶	P101	Laptop	Dell	ORD099	51	48
	P101	Laptop	Dell	ORD092	53	52
	P101	Laptop	Dell	ORD085	40	40
	P101	Laptop	Dell	ORD078	33	32
	P101	Laptop	Dell	ORD071	42	42

#UNION

24.Display all orders

with:OrderID,Product,Supplier,OrderedQty,DeliveredQty

```

select s.OrderID,
d.Product,
d.Supplier,
s.OrderedQty,
s.DeliveredQty
from supply_chain s
left join dim_material d
on s.ProductID = d.ProductID

```

UNION

#Display all products from dim_material, even if they have no orders.

```

SELECT
d.ProductID,
d.Product,

```

d.Supplier,

s.OrderID,

s.OrderedQty

FROM dim_material d

RIGHT JOIN supply_chain s

ON d.ProductID = s.ProductID;

Result Grid   Filter Rows: Export:  Wrap Cell Content: 					
	OrderID	Product	Supplier	OrderedQty	DeliveredQty
▶	ORD001	Laptop	Dell	45	45
	ORD002	Chair	IKEA	57	57
	ORD003	Mobile	Samsung	7	5
	ORD004	Printer	HP	48	45
	ORD005	Table	HomeCenter	14	14